

8 Space

9 Space II



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 8

Space

Tuesday 29 October 2024

Announcements

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■ Midterm test:

- *Date:* Tuesday 12 November 2024
- *Time:* 2:30pm–4:30pm
- *Location:* in class, HH-102
- Test structure will be discussed in class next week.

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- **Assignment 4** is due the day before the midterm.

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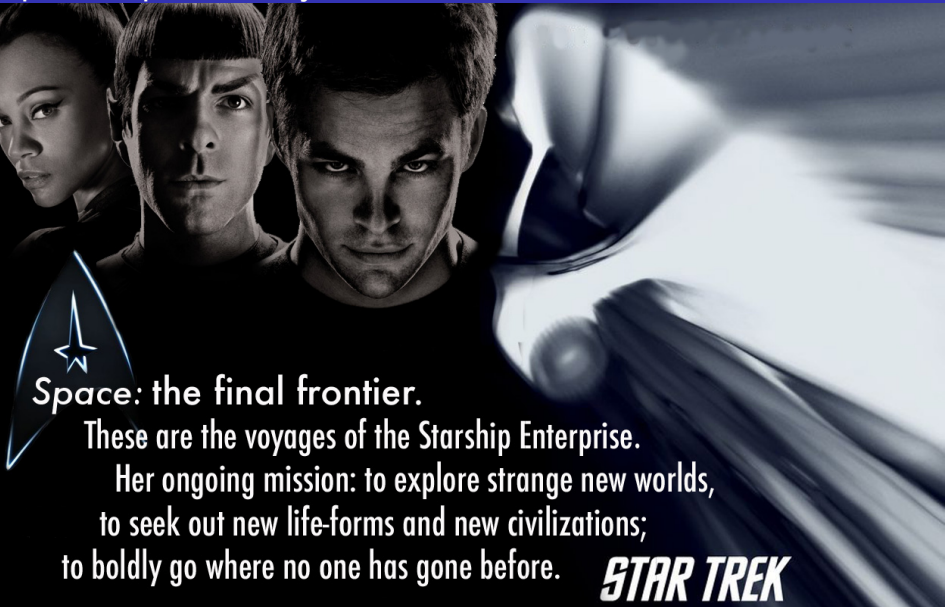
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- Make sure you personally can do the question on calculating $\mathcal{R}_0$ on this assignment before the midterm test.

Spatial Epidemic Dynamics

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Space: the final frontier.

These are the voyages of the Starship Enterprise.

Her ongoing mission: to explore strange new worlds,
to seek out new life-forms and new civilizations;
to boldly go where no one has gone before.

STAR TREK

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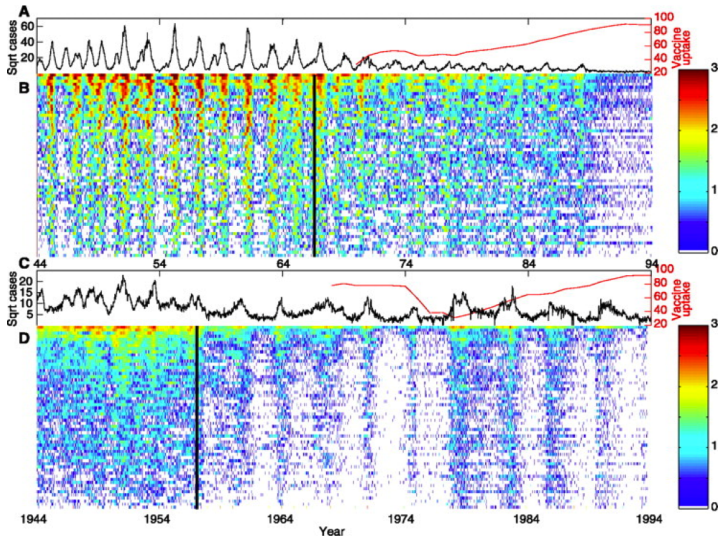
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- Can we reduce the eradication threshold below $p_{\text{crit}} = 1 - \frac{1}{\mathcal{R}_0}$?

Measles and Whooping Cough in 60 UK cities

Measles



Whooping
Cough

Rohani, Earn & Grenfell (1999) *Science* 286, 968–971

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- Can we manipulate epidemics predictably so as to increase probability of eradication?
- Can we eradicate measles?

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- Avoid spatially structured epidemics...
- Time to think about the mathematics of synchrony...
- But analytical theory of synchrony in a periodically forced system of differential equations is mathematically demanding...
- So let's consider a much simpler biological model...

The Logistic Map

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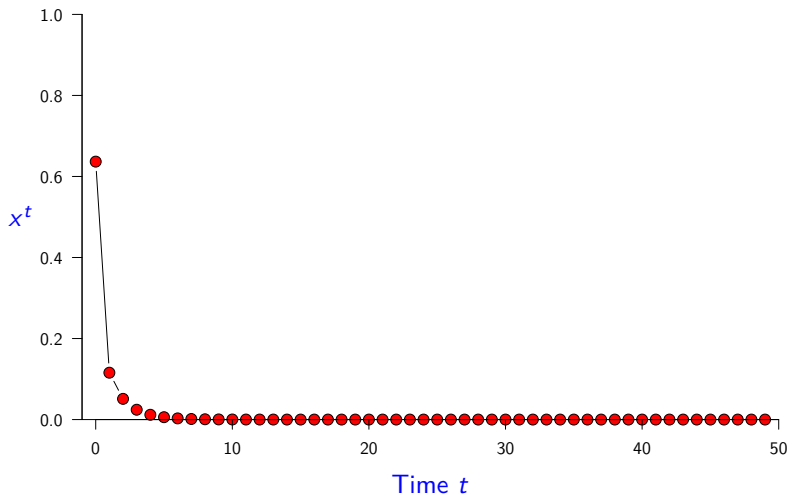
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- What kinds of dynamics are possible for the Logistic Map?

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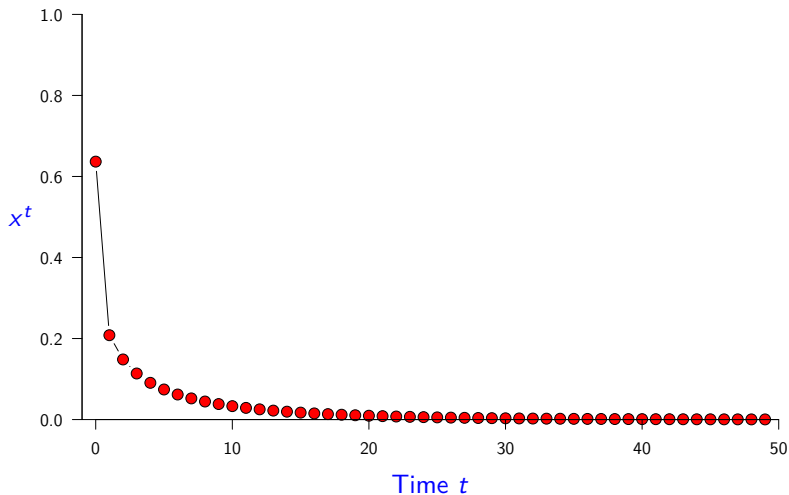
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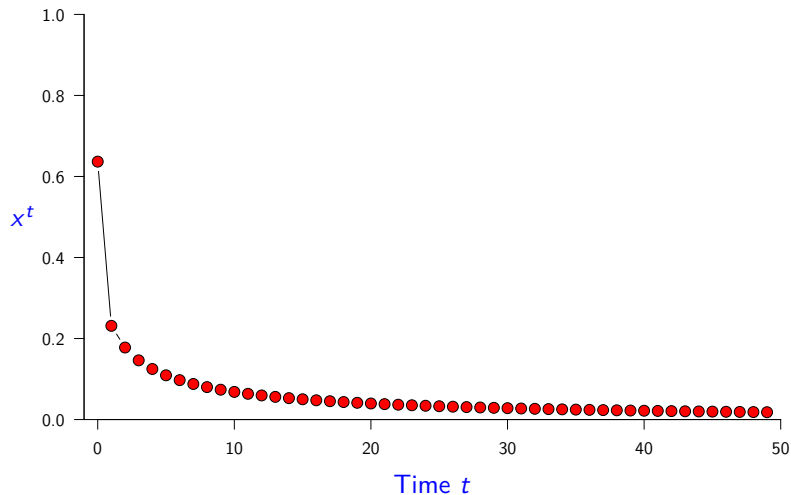
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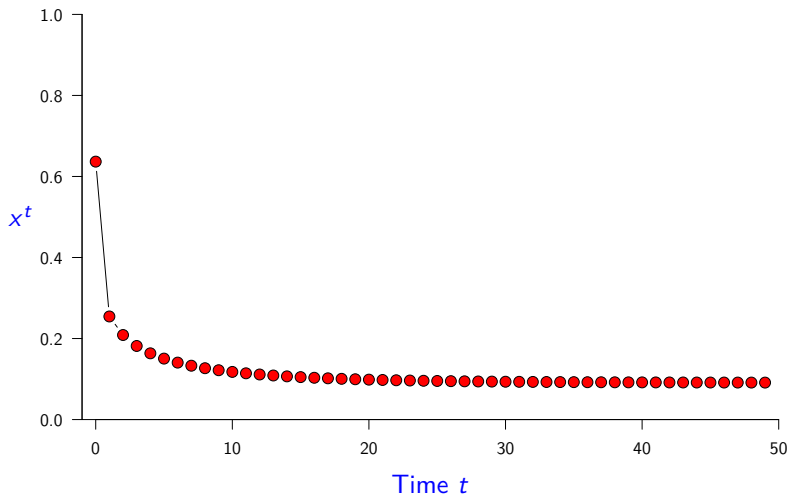
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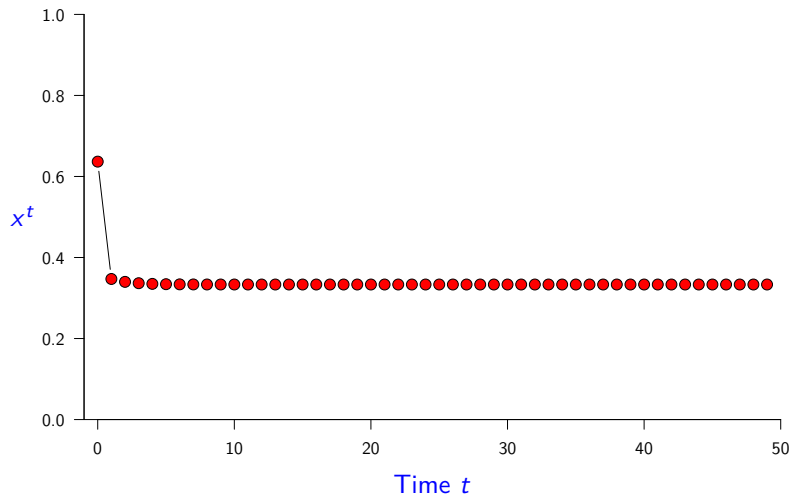
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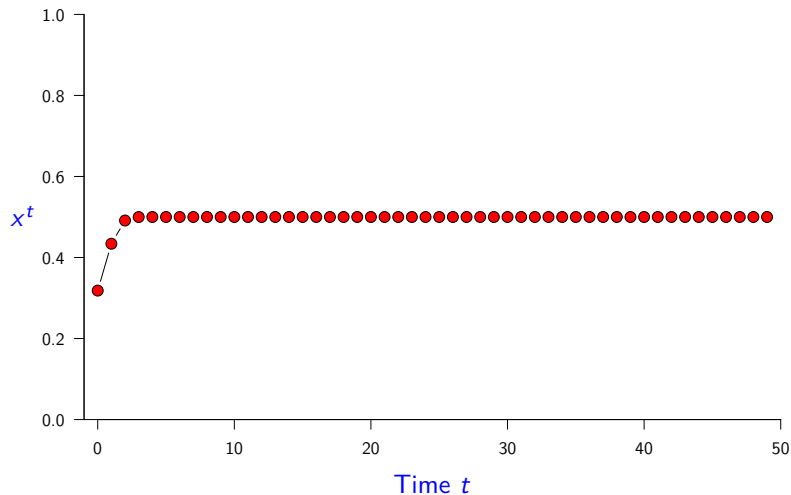
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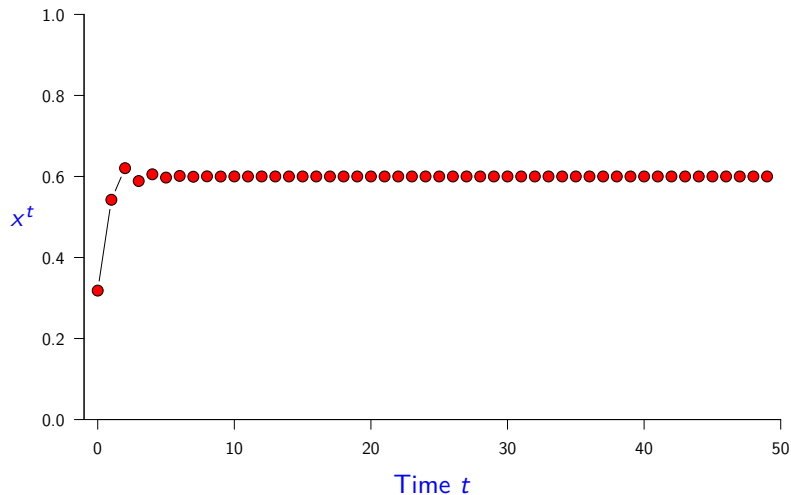
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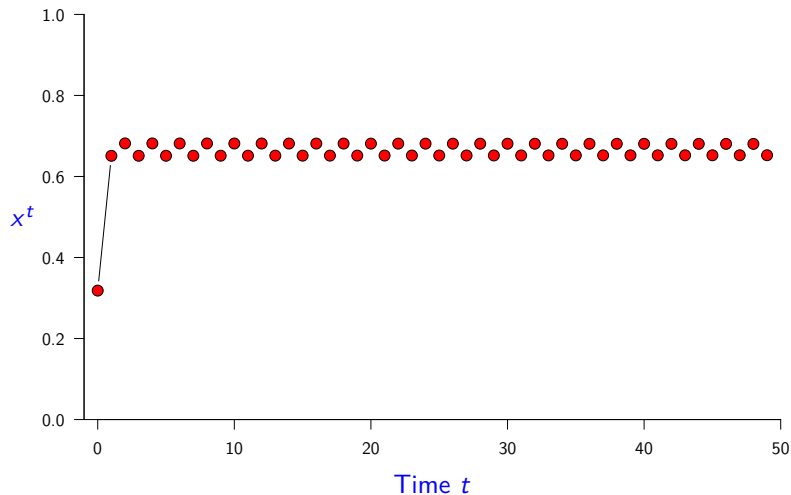
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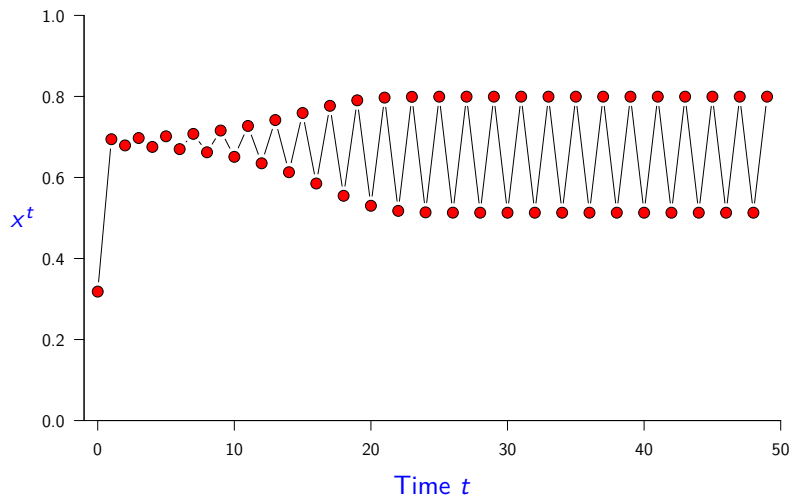
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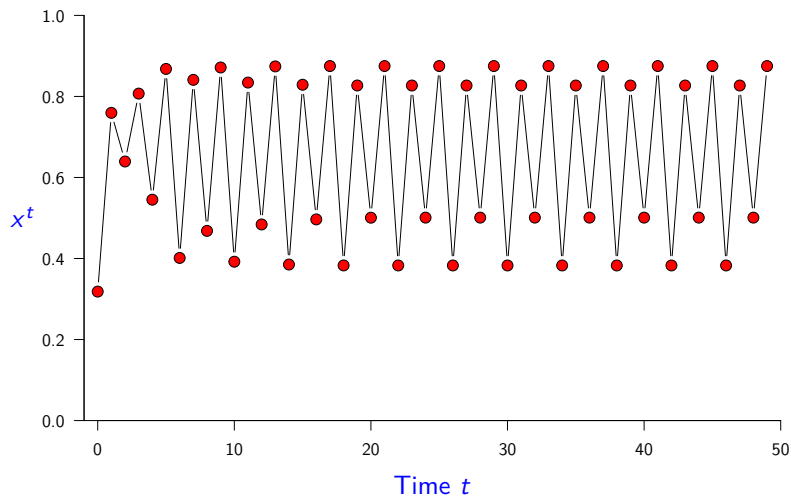
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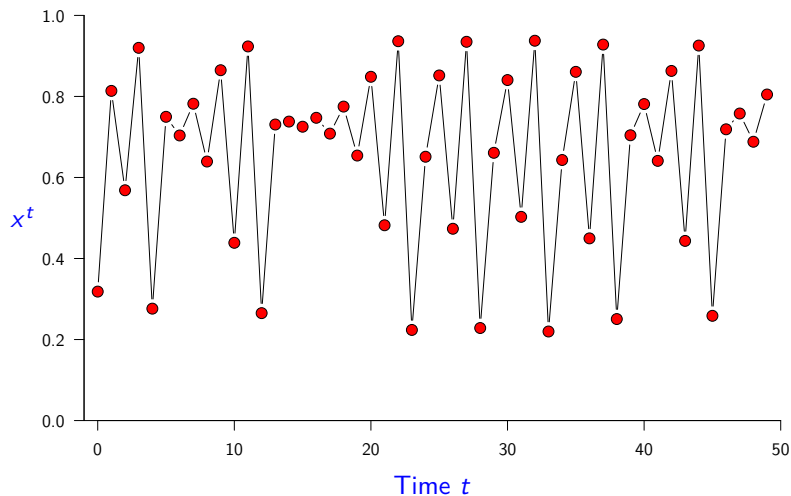
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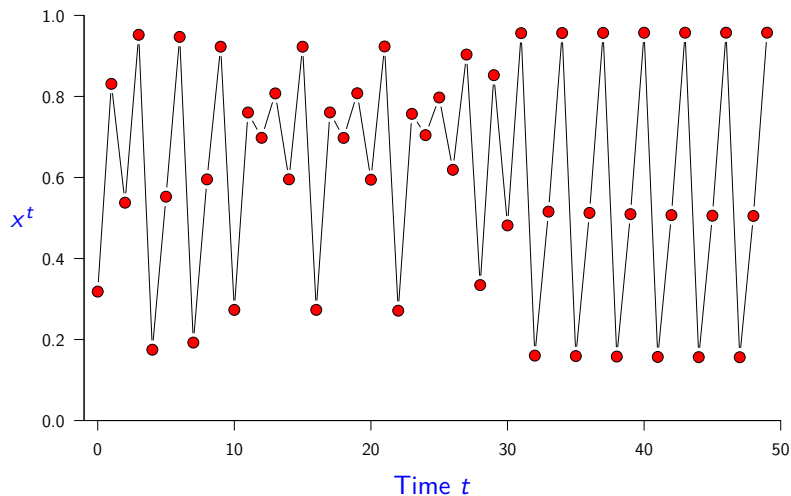
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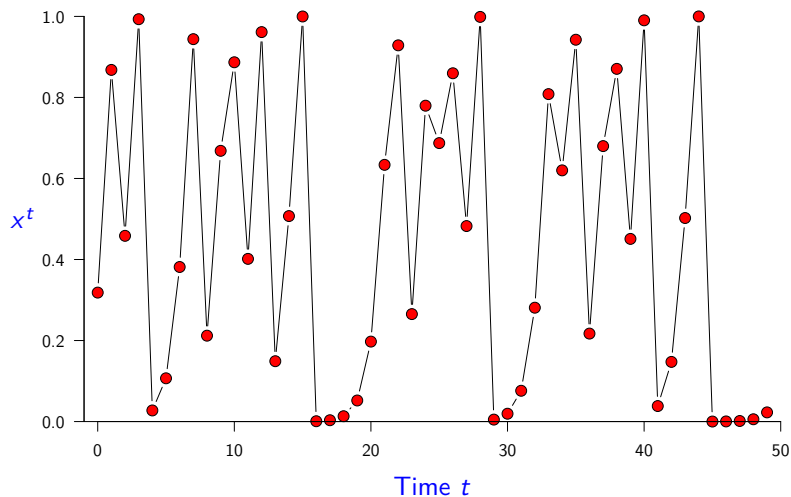
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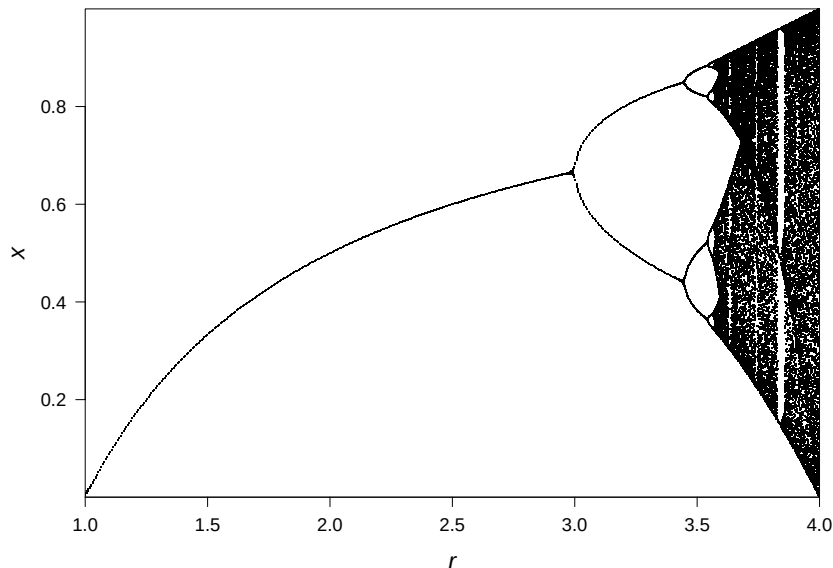
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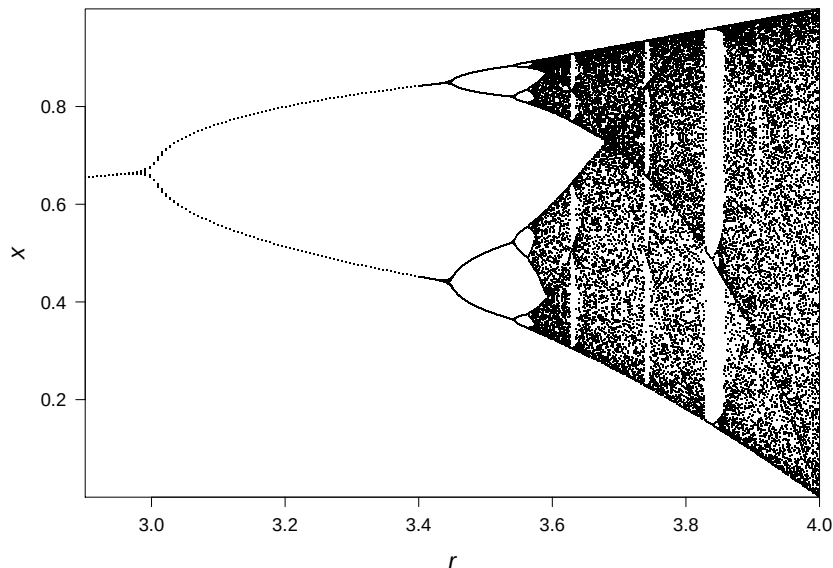
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 - Ignore transient behaviour: just show attractor.

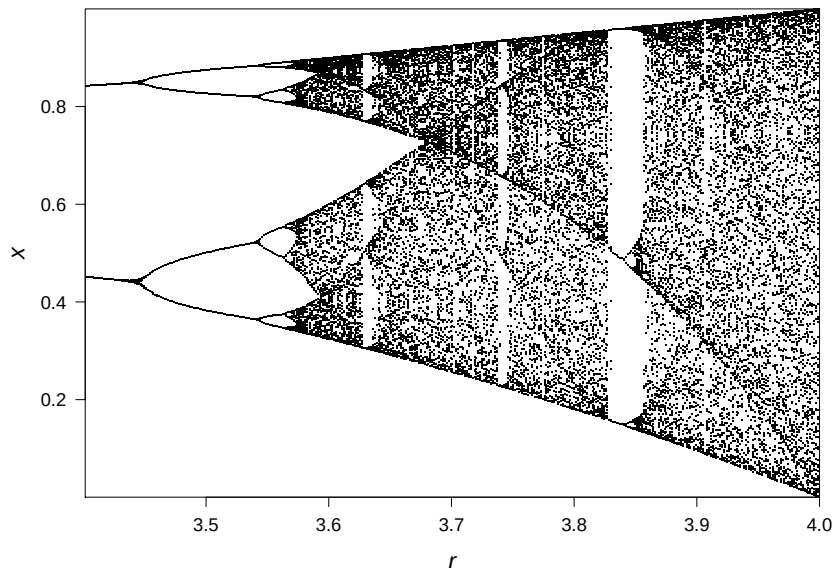
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- System is *coherent* (perfectly synchronous) if the state is the same in all patches
i.e., $\mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n$

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$$\text{i.e.,} \quad \begin{pmatrix} x_1^{t+1} \\ \vdots \\ x_n^{t+1} \end{pmatrix} = \begin{pmatrix} m_{11} & \cdots & m_{1n} \\ \vdots & \ddots & \vdots \\ m_{n1} & \cdots & m_{nn} \end{pmatrix} \begin{pmatrix} F(x_1^t) \\ \vdots \\ F(x_n^t) \end{pmatrix}$$

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- *Colour coding of matrix indices:*
 - row indices are red
 - column indices are cyan

Basic properties of dispersal matrices $M = (m_{ij})$

Discrete-time *metapopulation* model:

$$x_{\textcolor{red}{i}}^{t+1} = \sum_{\textcolor{blue}{j}=1}^n m_{\textcolor{blue}{i}\textcolor{blue}{j}} F(x_{\textcolor{blue}{j}}^t), \quad \textcolor{red}{i} = 1, 2, \dots, n.$$

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(*each m_{ij} is non-negative and at most 1*)
- Total proportion that leaves or stays in patch j :

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so any coherent state can be written $x\mathbf{e}$, for some $x \in \mathbb{R}$.

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Back to Space and Synchrony

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 - If so, then this sort of analysis should help to identify synchronizing vaccination strategies.

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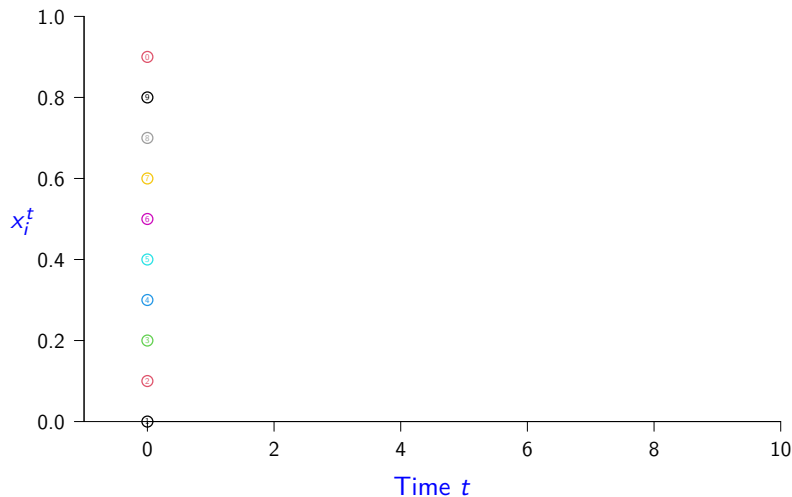
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Logistic Metapopulation Simulation ($r = 1$, $m = 0.2$)

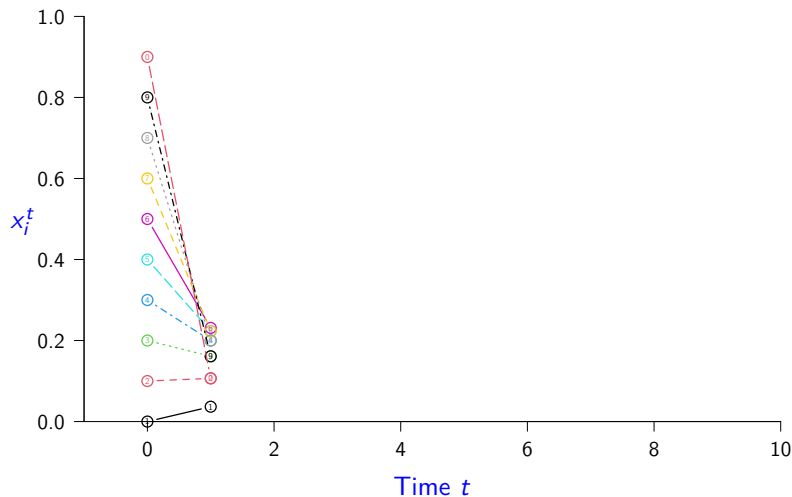
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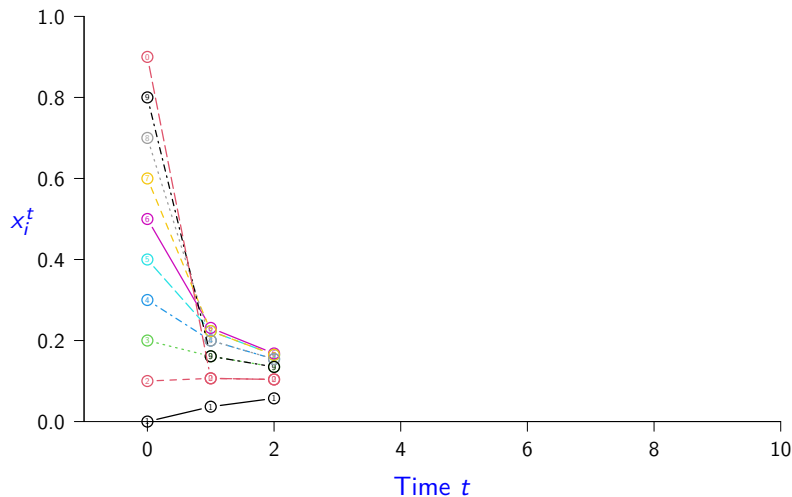
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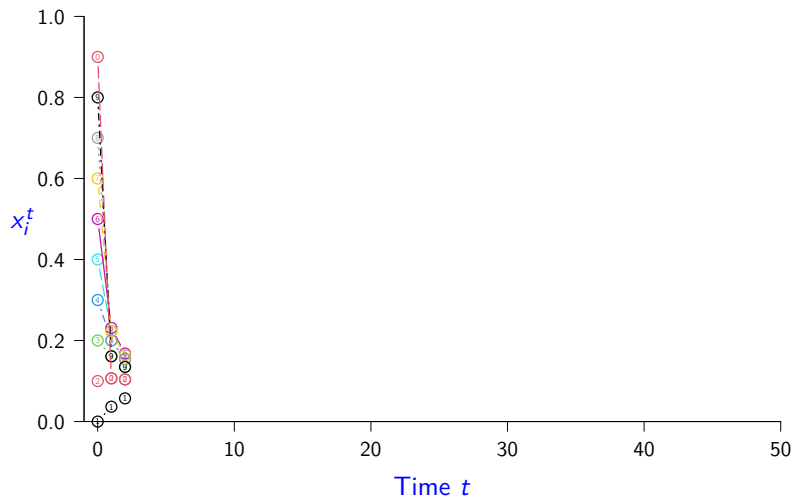
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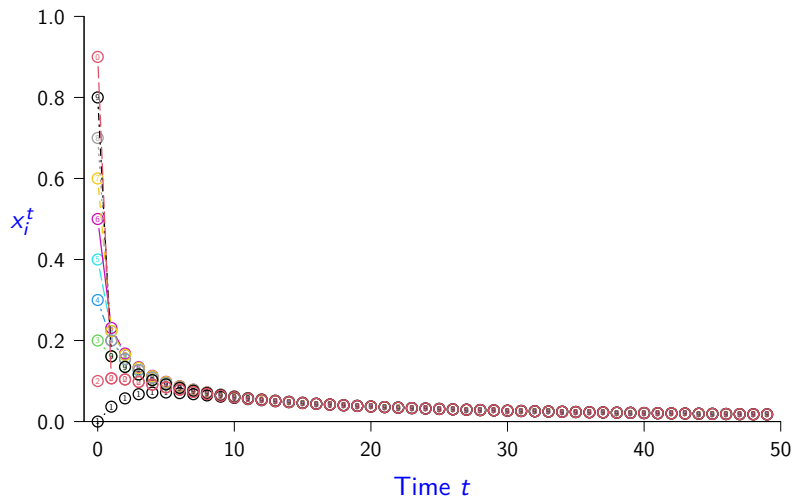
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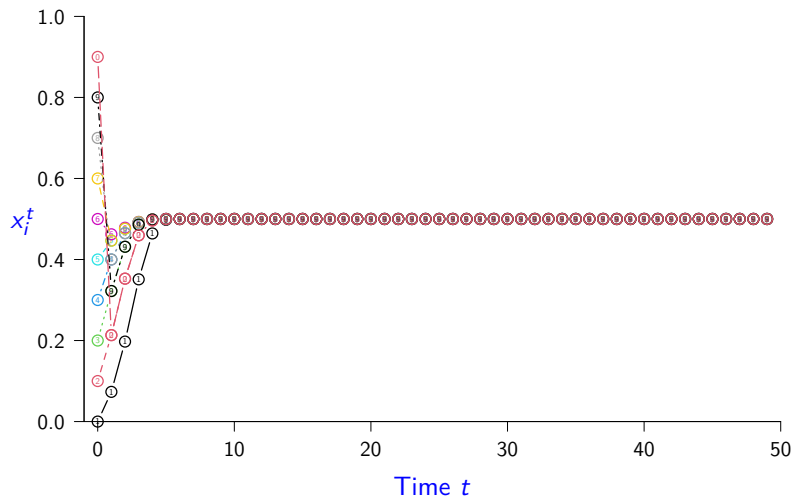
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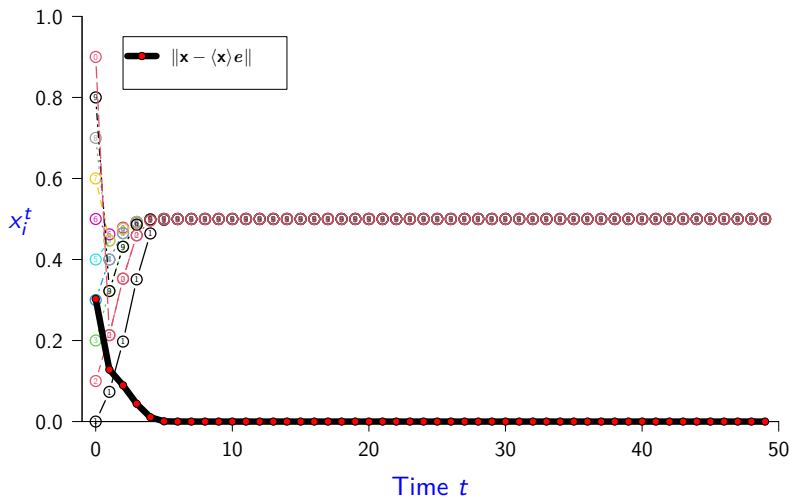
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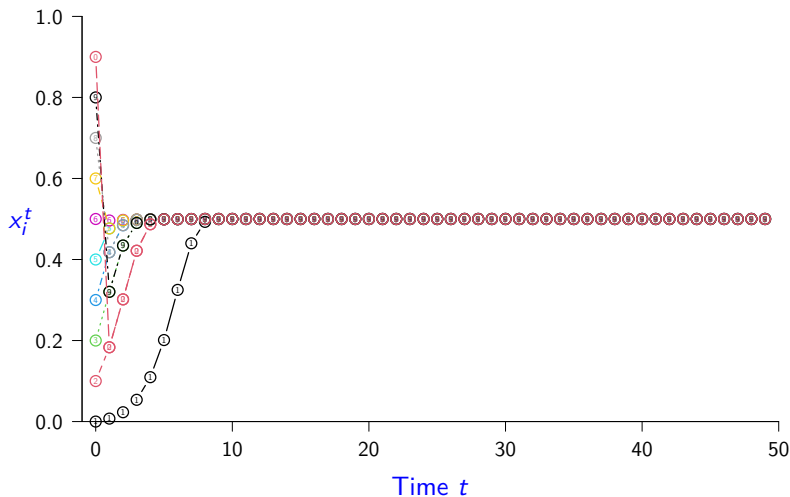
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Logistic Metapopulation Simulation ($r = 2$, $m = 0.02$)

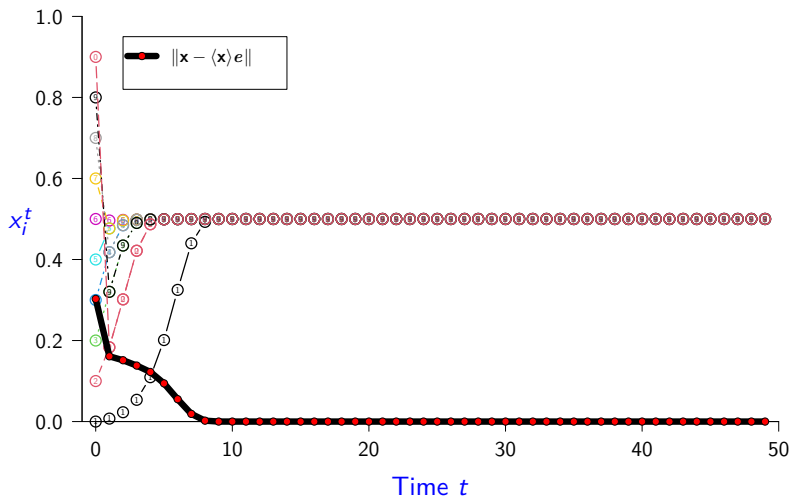
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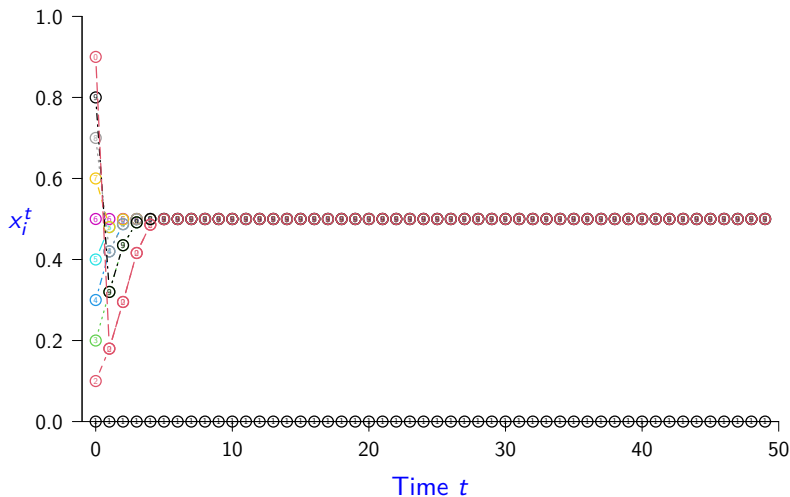
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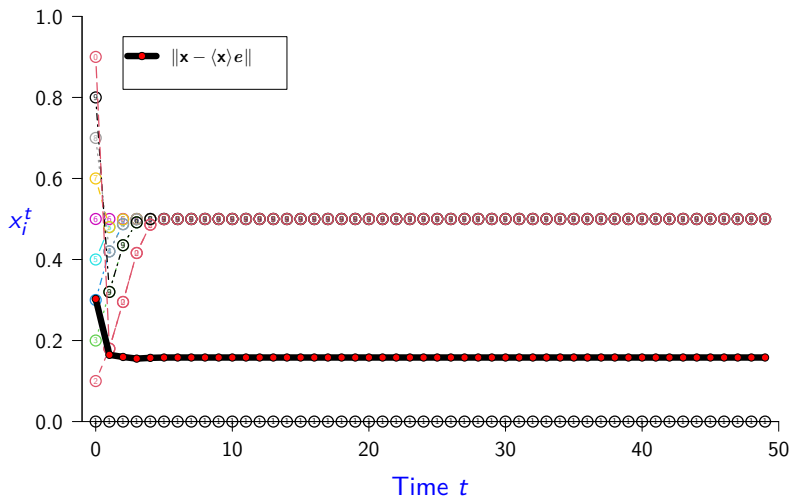
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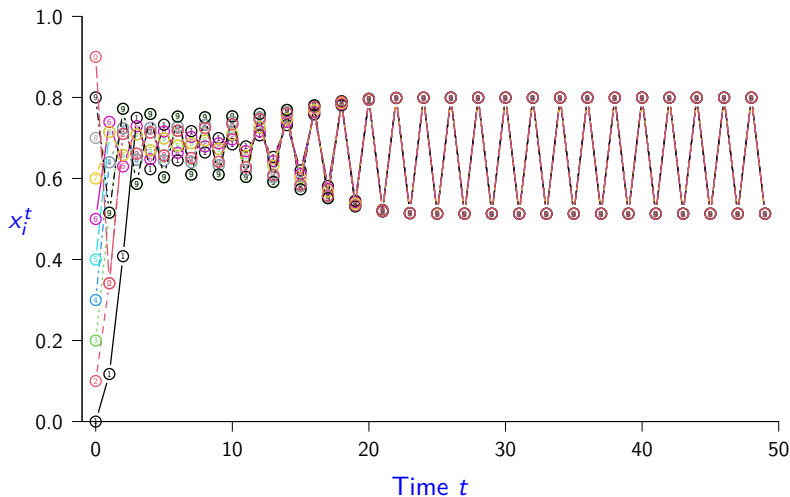
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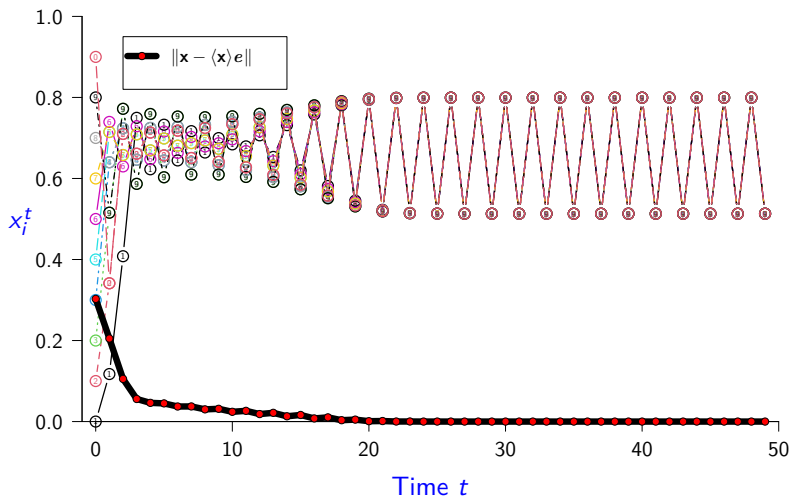
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Logistic Metapopulation Simulation ($r = 3.2$, $m = 0.2$)

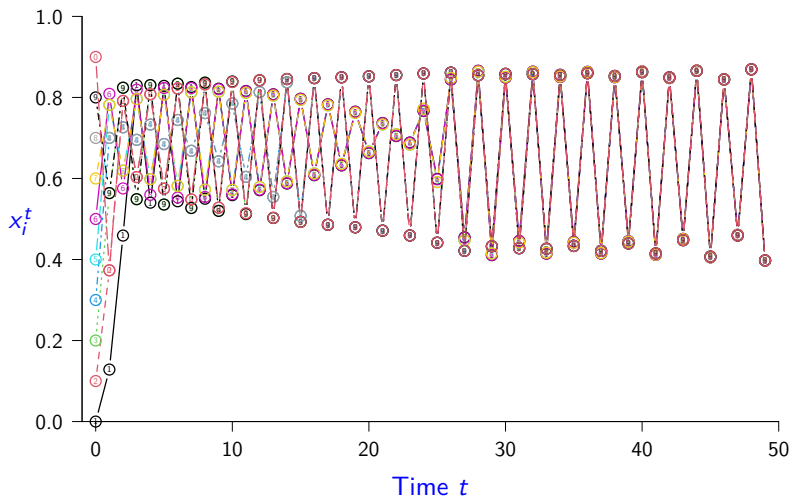
$n = 10$, $r = 3.2$, $m = 0.2$, $\lambda = 0.778$



Logistic Metapopulation Simulation ($r = 3.5$, $m = 0.2$)

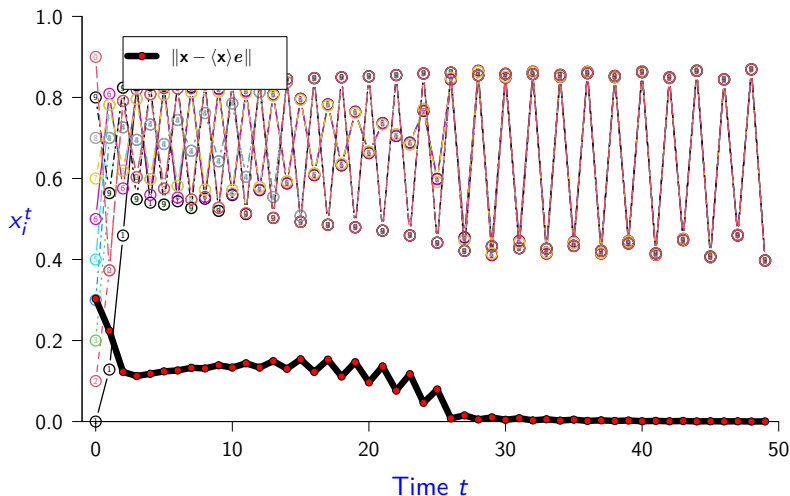
Logistic Metapopulation Simulation ($r = 3.5$, $m = 0.2$)

$n = 10$, $r = 3.5$, $m = 0.2$, $\lambda = 0.778$



Logistic Metapopulation Simulation ($r = 3.5$, $m = 0.2$)

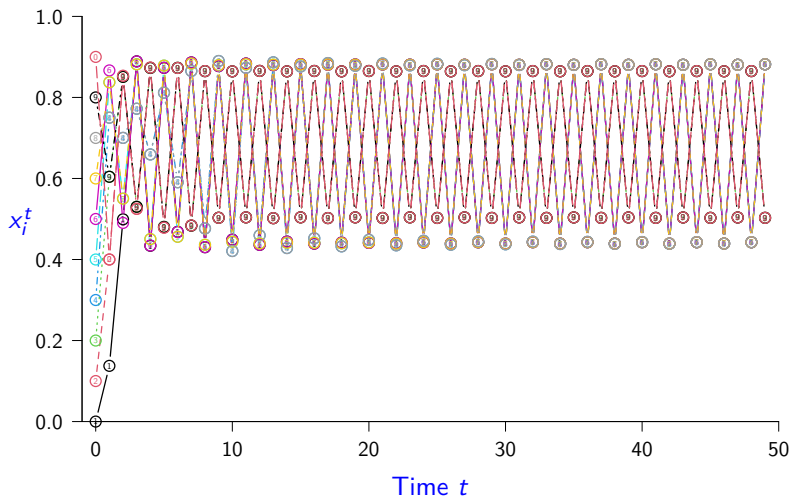
$$n = 10, \quad r = 3.5, \quad m = 0.2, \quad \lambda = 0.778$$



Logistic Metapopulation Simulation ($r = 3.75$, $m = 0.2$)

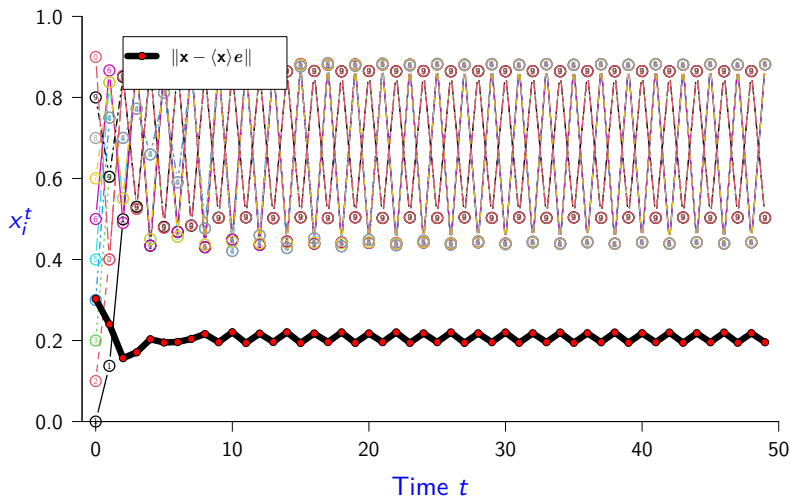
Logistic Metapopulation Simulation ($r = 3.75$, $m = 0.2$)

$n = 10$, $r = 3.75$, $m = 0.2$, $\lambda = 0.778$



Logistic Metapopulation Simulation ($r = 3.75$, $m = 0.2$)

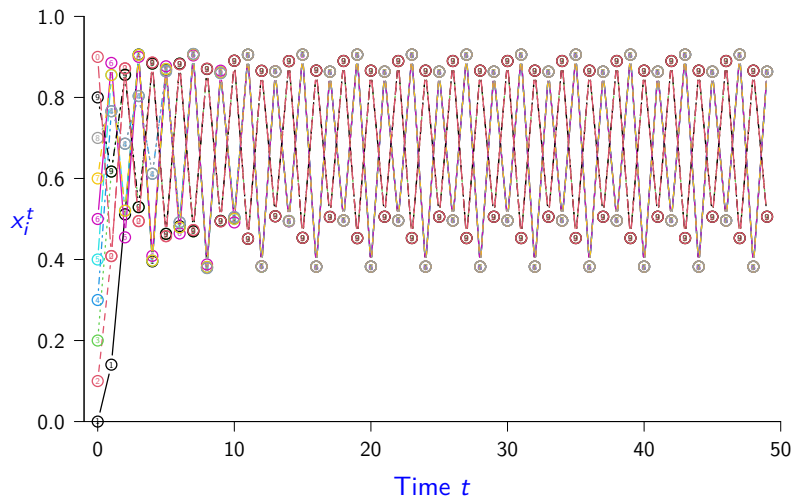
$$n = 10, \quad r = 3.75, \quad m = 0.2, \quad \lambda = 0.778$$



Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.2$)

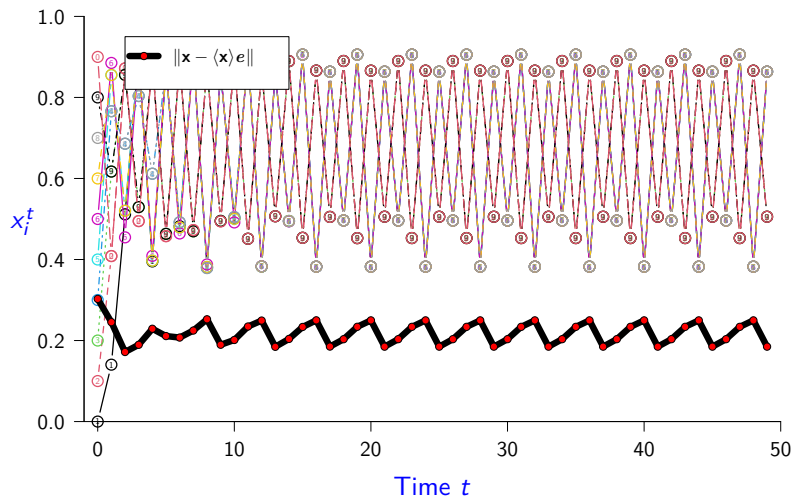
Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.2$)

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Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.2$)

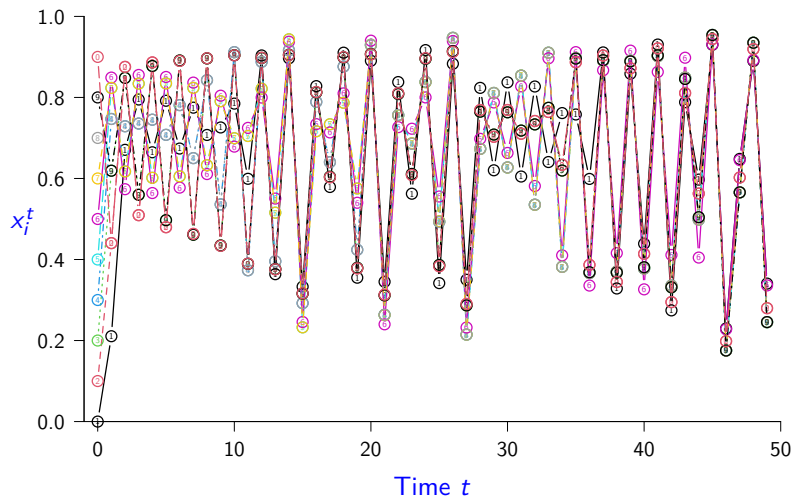
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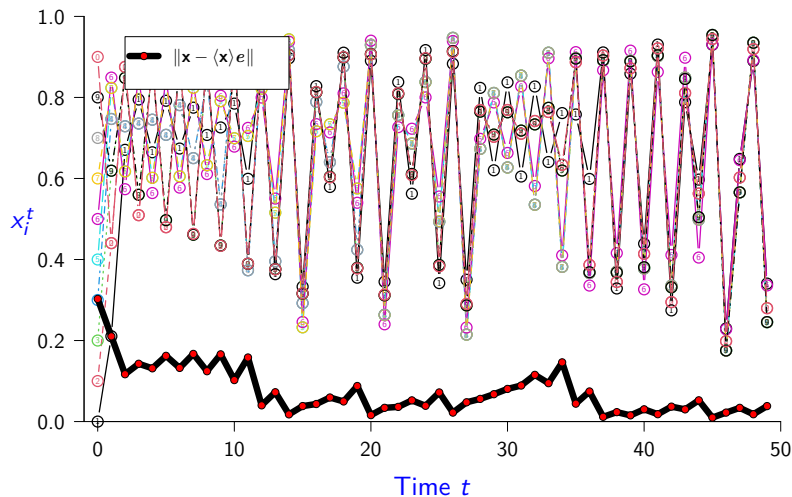
Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.3$)

$n = 10$, $r = 3.83$, $m = 0.3$, $\lambda = 0.667$



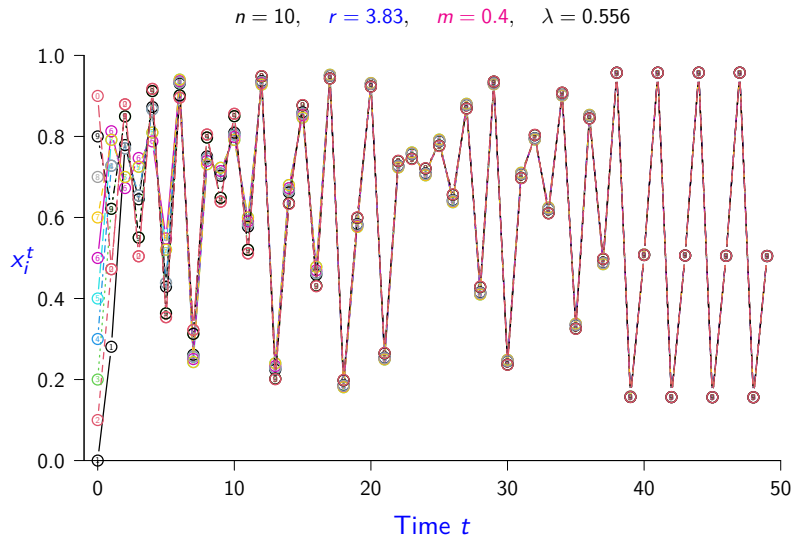
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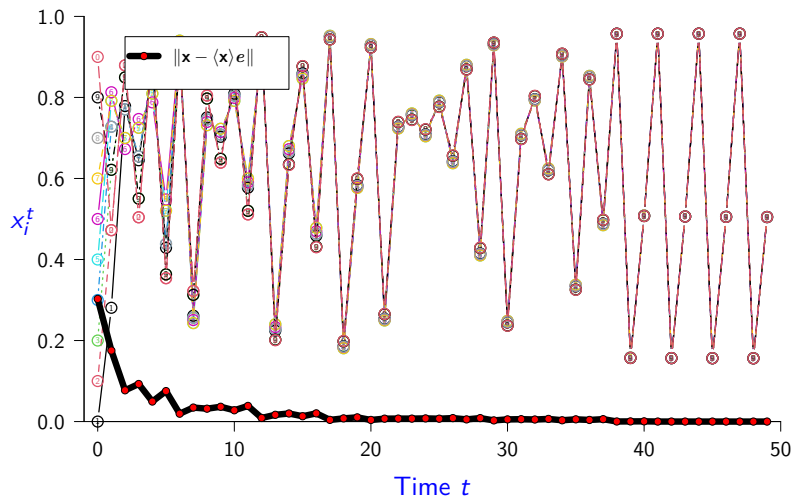
Logistic Metapopulation Simulation ($r = 3.83$, $m = 0.4$)

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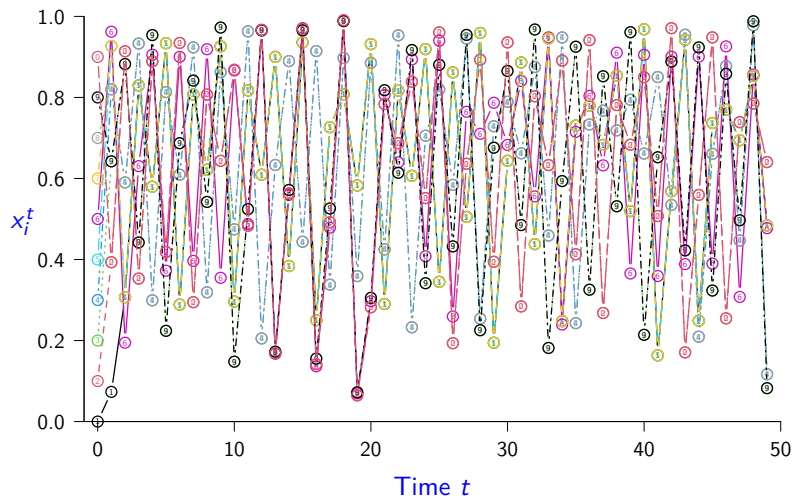
$n = 10$, $r = 3.83$, $m = 0.4$, $\lambda = 0.556$



Logistic Metapopulation Simulation ($r = 4$, $m = 0.1$)

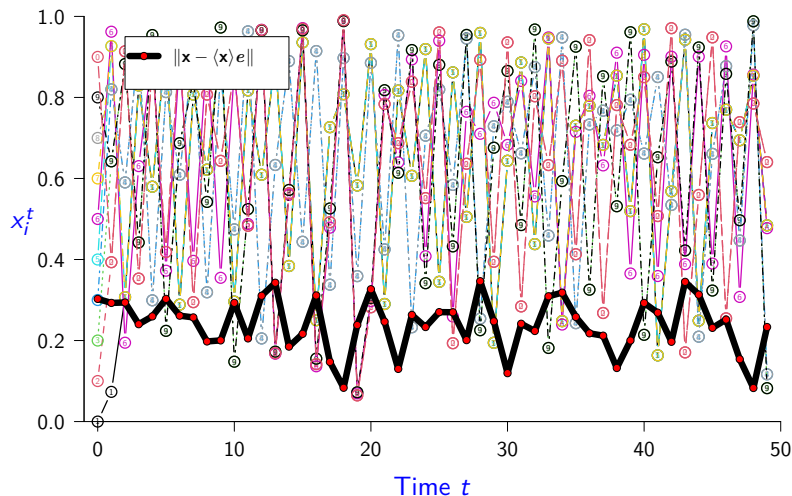
Logistic Metapopulation Simulation ($r = 4, m = 0.1$)

$n = 10, \quad r = 4, \quad m = 0.1, \quad \lambda = 0.889$



Logistic Metapopulation Simulation ($r = 4, m = 0.1$)

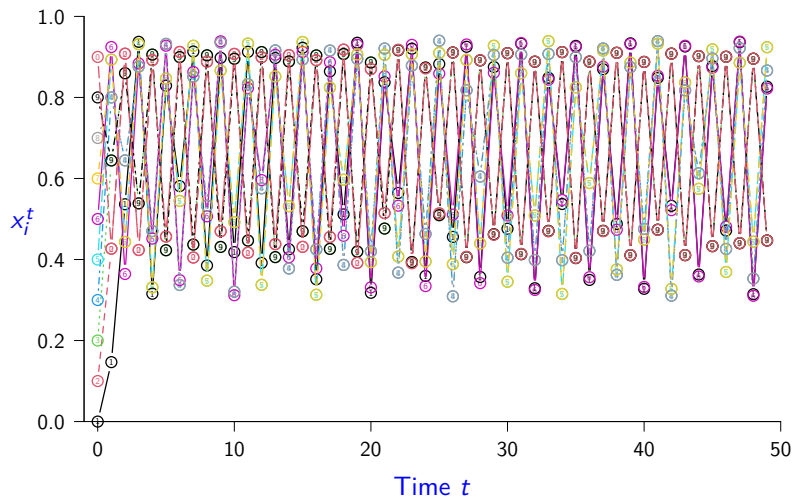
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Logistic Metapopulation Simulation ($r = 4$, $m = 0.2$)

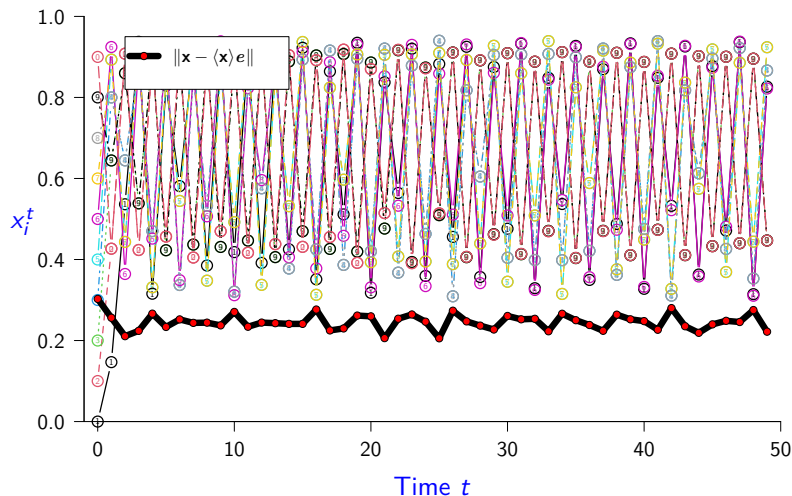
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Logistic Metapopulation Simulation ($r = 4, m = 0.2$)

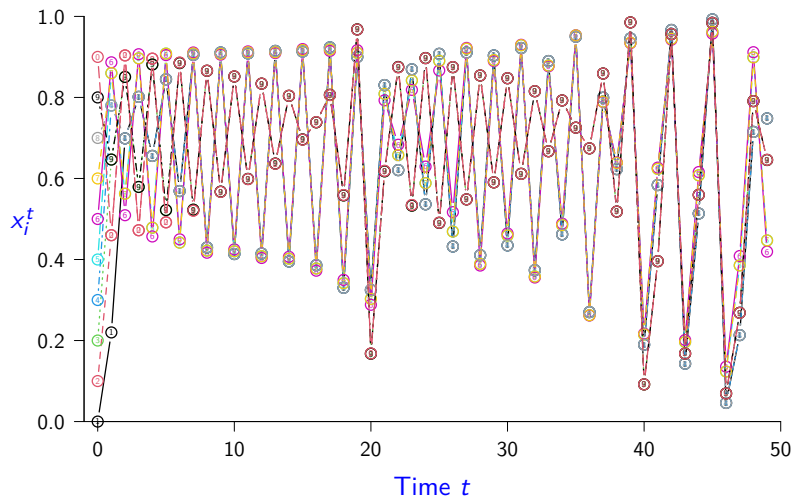
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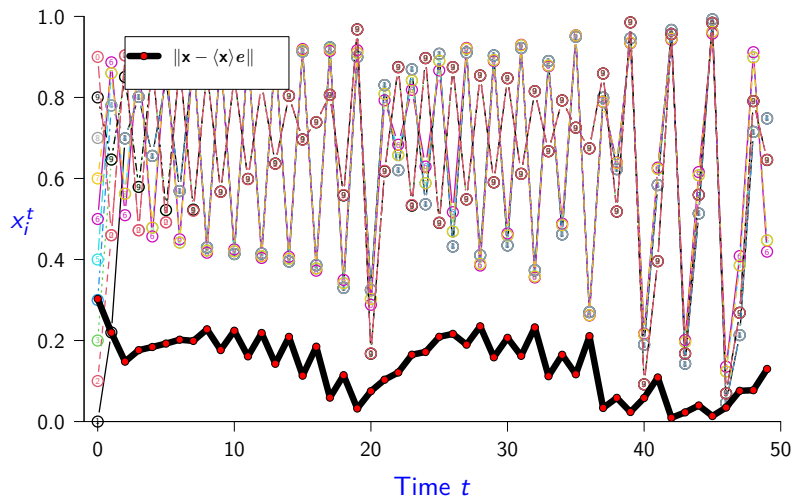
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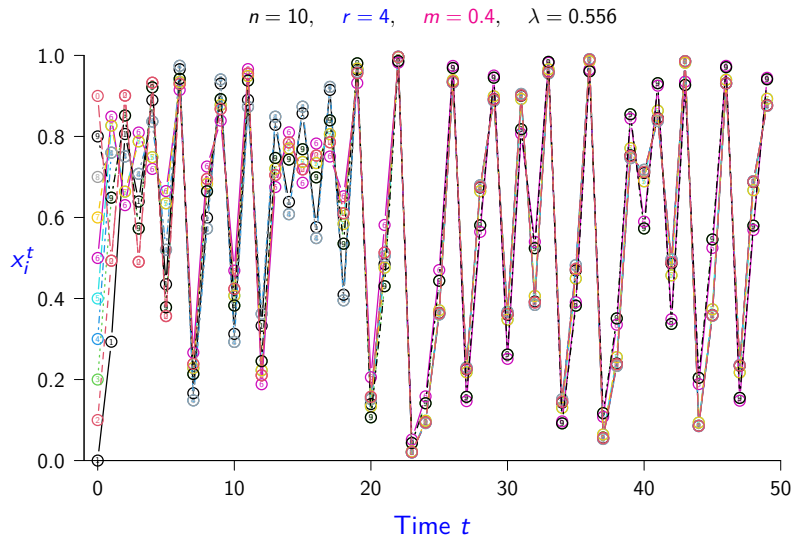
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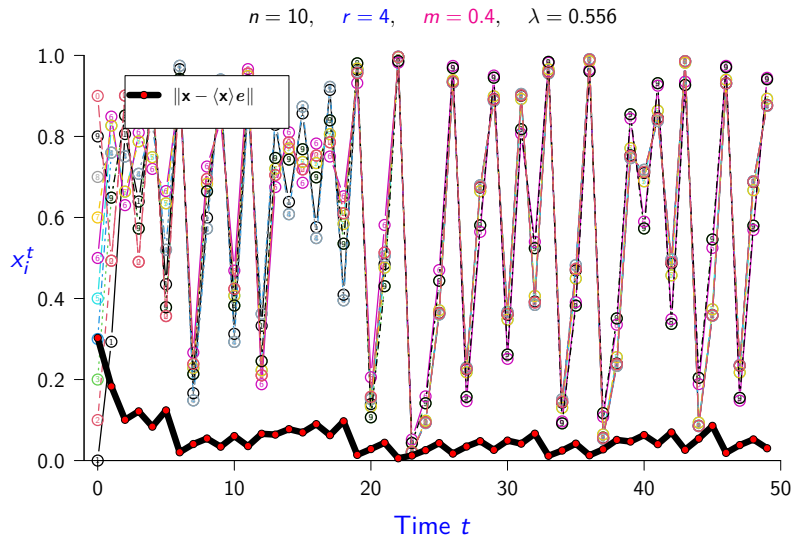


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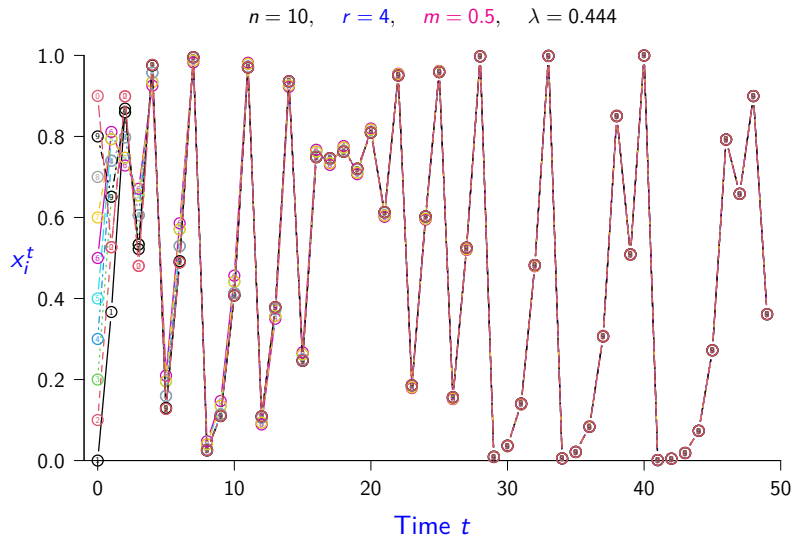


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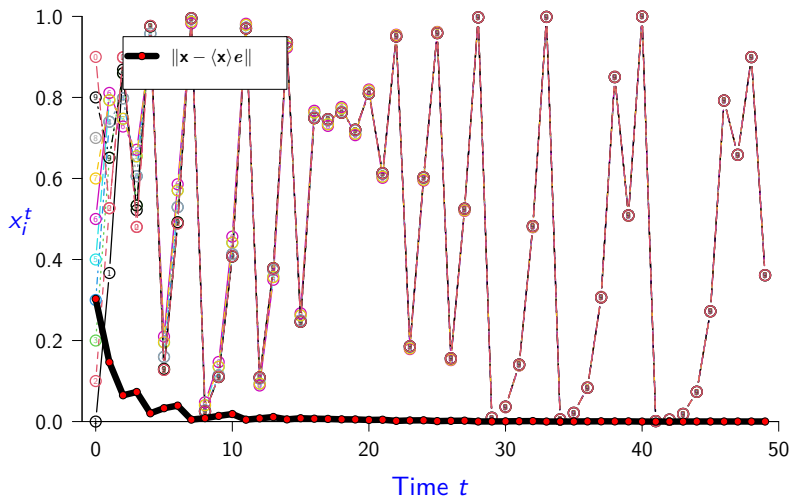
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- Logistic Metapopulation Simulations (10 patches)

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- $r = 2, m = 0.2$

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- Coherence is affected by magnitude $|\lambda|$ of *subdominant eigenvalue* λ .

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A vector is *positive* if each of its components is positive.

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Proof.

See Horn & Johnson (2013) *Matrix Analysis*, Corollary 8.1.30, p. 522. $\square$

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Consider a discrete-time metapopulation map,

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But e is a positive vector, hence by the lemma on the previous slide, 1 is a dominant eigenvalue of M . □

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- We actually want the average over the entire trajectory, so we would like to consider

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(true for us because we assume $r = \max_x \|D_x F\|$ exists).

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This is called the maximum (Lyapunov) *characteristic exponent* of the single patch map.

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Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 4MB3/6MB3 Mathematical Biology

Instructor: David Earn

Lecture 9

Space II

Tuesday 5 November 2024

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 - The relationship between the rate of leaving a compartment and the mean time spent in the compartment.
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 - Finding equilibria of discrete time models, e.g., models of the form $x^{t+1} = F(x^t)$.

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 - Make sure you understand and can explain bifurcation diagrams with respect to seasonal amplitude (α) and with respect to basic reproduction number ($\mathcal{R}_0$). In particular, make sure you can explain how relevant bifurcation diagrams can be used to explain transitions in dynamics of infectious diseases that cause recurrent epidemics.

Midterm Test

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4mb-6mb-test-cb386

#1 Page 1 of 16



MATHEMATICS 4MB3/6MB3

Midterm Test, Tuesday 12 November 2024

First name (please write as legibly as possible within the boxes)

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Last name

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Student ID number

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Special Instructions and Notes:

- (i) This test has **16** pages. Verify that your copy is complete. Note that the final three pages are blank to provide additional space if needed.
- (ii) **Answer all questions in the space provided.**
- (iii) It is possible to obtain a total of 100 marks. There are 8 multiple choice questions (question 1 is worth 2 marks, whereas all other multiple choice questions are worth 4 marks each). There are 10 short answer questions (worth 7 marks each).
- (iv) **For multiple choice questions, circle only one answer.**
- (v) No calculators, notes, or aids of any kind are permitted.
- (vi) PHAC refers to the Public Health Agency of Canada.

GOOD LUCK

Coherence: what we've seen so far

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Global asymptotic coherence (GAC) for equal coupling

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Note: Actually true for very general connectivity matrices M and multi-dimensional single-patch dynamics $F(\mathbf{x})$.

Earn & Levin (2006) *PNAS* **103**, 3968-3971

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- Quasi-global theory attempts to bridge the gap between “probability = 1” and “probability > 0 ”.

McCluskey & Earn (2011) *J. Math. Biol.* **62**, 509–541

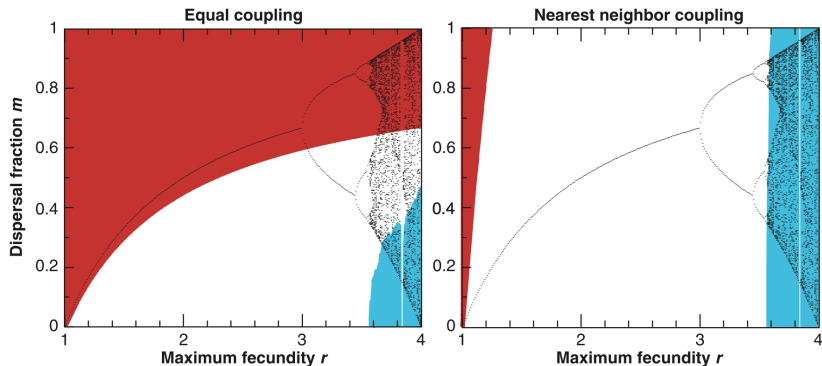
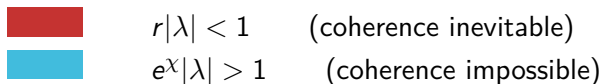
Application of simple coherence criteria

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10 patch logistic metapopulation

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Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

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McCluskey & Earn (2011) *J. Math. Biol.* 62, 509-541

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10 patch logistic metapopulation

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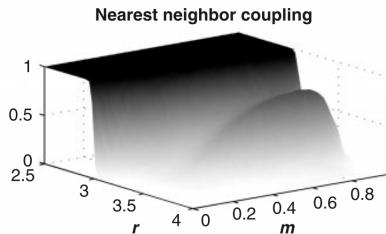
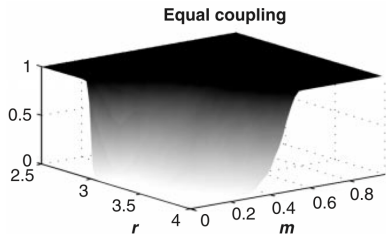
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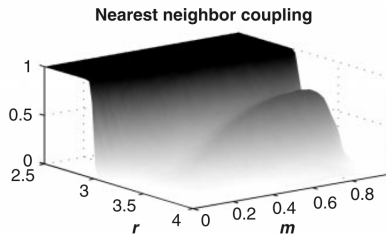
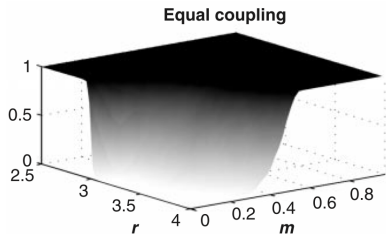


Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

Coherence in “numerical experiments” (simulations)

10 patch logistic metapopulation

- Systematically explore representative set of initial conditions and determine probability of coherence within some tolerance, within some specified time
 - e.g., coherence to within 10% within 10 iterations



Earn, Levin & Rohani (2000) *Science* **290**, 1360–1364

- *Extremely demanding computationally. . .*

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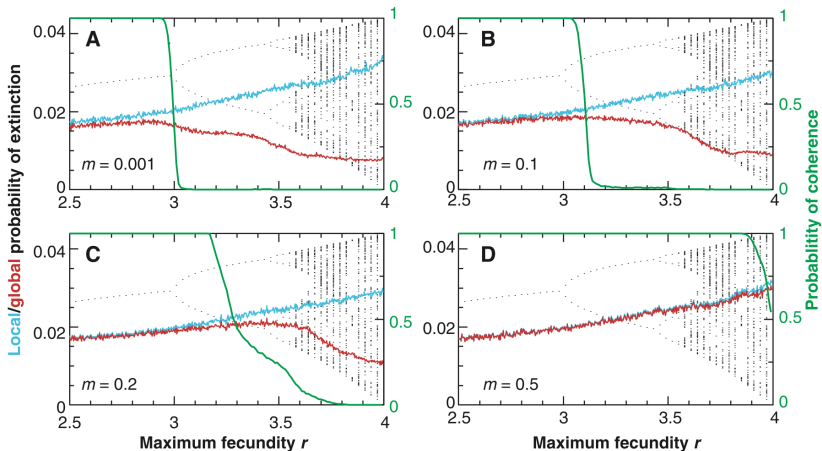
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- Coherence criteria yield method for estimating *risk of synchronization* in ecological systems

Earn, Levin & Rohani (2000) "Coherence and Conservation" *Science* **290**, 1360–1364

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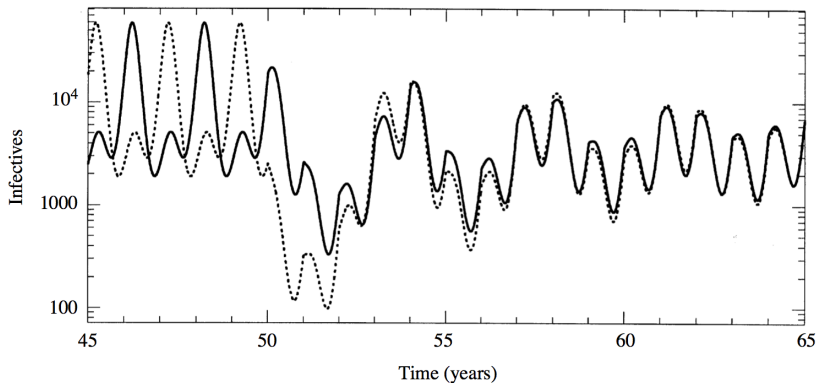
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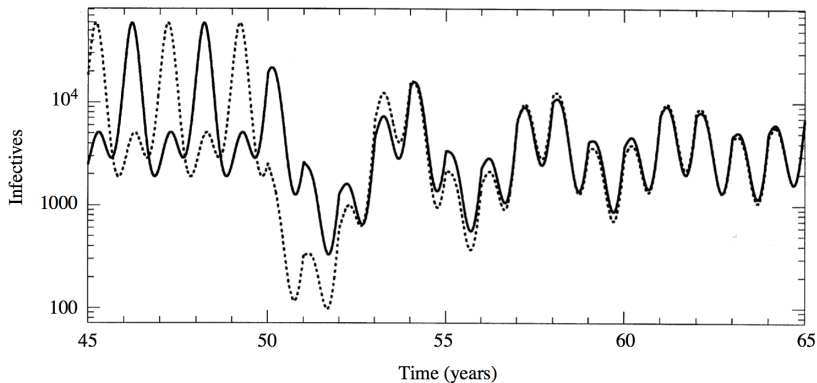
- Periodic forcing can have complex dynamical effects. . .

Example of Synchronization via Pulse Vaccination



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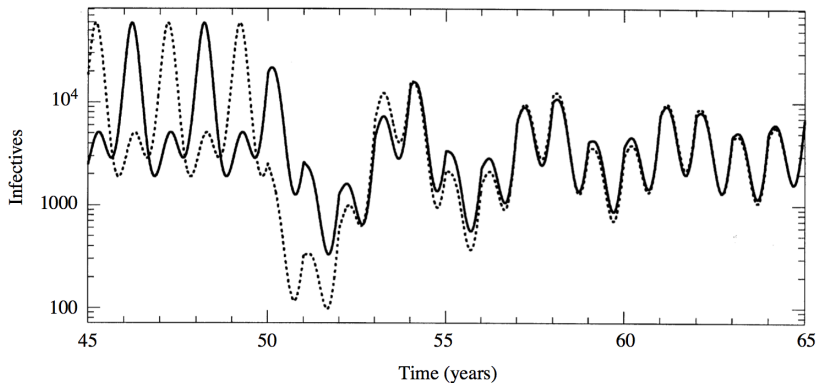
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Earn, Rohani & Grenfell (1998) *Proc. R. Soc. Lond.* **265**, 7–10